STAT A411/611 Lecture Notes

Instructor: Rieken Venema **Office:** SSB 158E
Office hours: MWF 12:00–12:45 **Class:** 2:30–3:45

Today: Sections 4.4 and 4.5.

Wednesday: Section 5.1.

Chapter 4: Continuous Random Variables

A continuous random variable X takes values in an interval, for example

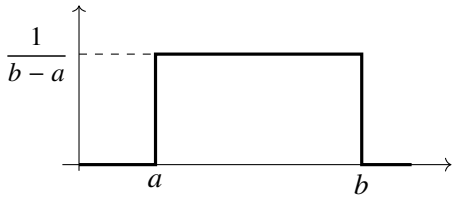
$$[0, \infty), \qquad [0, 1], \qquad (-\infty, \infty).$$

Its probability distribution is described by a probability density function (pdf) $p(x)$. The cumulative distribution function (cdf) is

$$F(x) = P(X \le x) = \int_{-\infty}^{x} p(t)\, dt, \qquad F'(x) = p(x).$$

$$p(x) = 12.5x - 1.25.$$

Uniform Distribution: $X \sim U(a, b)$



$$p(x) = \begin{cases} \dfrac{1}{b-a}, & a < x < b, \\ 0, & \text{otherwise.} \end{cases}$$

Section 4.2: Expected Values

For a function h of the continuous random variable,

$$E[h(X)] = \int_{-\infty}^{\infty} h(x)p(x)\, dx.$$

Related quantities and properties:

$$\mu = E[X], \qquad \sigma = \mathrm{sd}(X), \qquad E[X^k], \qquad E[aX + b] = aE[X] + b.$$

$$p(x) = \begin{cases} e^{-x}, & x > 0, \\ 0, & x \le 0. \end{cases}$$

Section 4.3: Gamma Function and Gamma Distribution

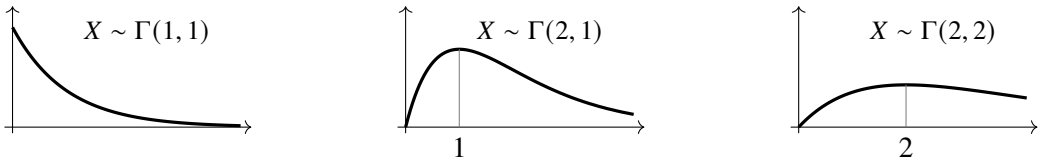
Gamma function:

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x}\, dx.$$

Gamma distribution: If $X \sim \Gamma(\alpha, \beta)$,

$$p(x) = \begin{cases} \dfrac{x^{\alpha-1} e^{-x/\beta}}{\beta^\alpha \Gamma(\alpha)}, & x > 0, \\ 0, & x \le 0, \end{cases} \qquad \alpha > 0,\ \beta > 0.$$

Examples of gamma density curves:



Two special cases are the exponential and chi-squared distributions:

$$X \sim \mathrm{E}(\beta) = \Gamma(1, \beta) \quad \text{(exponential)},$$
$$X \sim \chi^2_\nu = \Gamma\left(\frac{\nu}{2}, 2\right) \quad \text{(chi-squared)}.$$

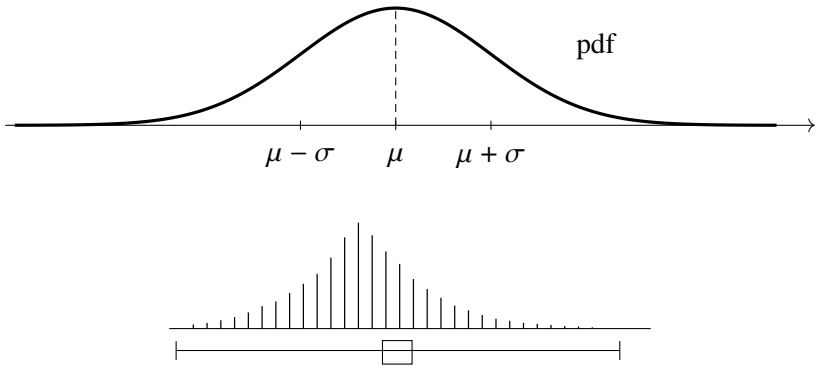
Section 4.4: The Normal Distribution

Normal PDF and Parameters

$$X \sim N(\mu, \sigma^2), \qquad \mu = E[X], \qquad \sigma = \mathrm{sd}(X).$$

Its probability density function is

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right], \qquad -\infty < x < \infty.$$



Standard Normal Distribution

X has the standard normal distribution when

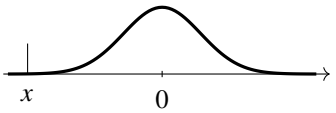
$$X \sim N(0,1), \qquad p(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \quad -\infty < x < \infty.$$

Standardizing (the Z-score):

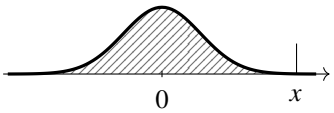
$$X \sim N(\mu, \sigma^2) \quad \Longrightarrow \quad Z = \frac{X-\mu}{\sigma} \sim N(0,1).$$

Table V: Standard Normal CDF

$$X \sim N(0,1), \qquad F(x) = P(X \le x).$$



$P(X \le x)$
$x = -3.49$

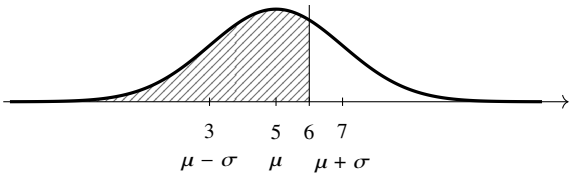


$P(X \le x)$
$x = 3.49$

Example: $X \sim N(5,4)$

Find $P(X \le 6)$.

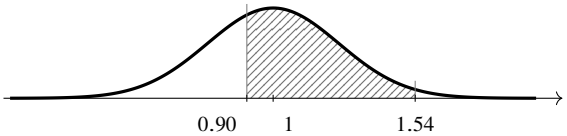
$$\mu = 5, \quad \sigma = 2, \quad \mu - \sigma = 3, \quad \mu + \sigma = 7.$$



$$\begin{aligned} P(X \le 6) &= P\left(\frac{X-5}{2} \le \frac{6-5}{2}\right) \\ &= P(Z \le 0.50) \qquad \left(Z = \frac{X-5}{2} \sim N(0,1)\right) \\ &= 0.6915. \end{aligned}$$

Example 4.4.2: $X \sim N\left(1, (0.25)^2\right)$

Find $P(0.90 < X < 1.54)$.



$$\begin{aligned} P(0.90 < X < 1.54) &= P(X \le 1.54) - P(X \le 0.90) \\ &= P\left(\underbrace{\frac{X-1}{0.25}}_{Z} \le \frac{1.54-1}{0.25}\right) - P\left(\frac{X-1}{0.25} \le \frac{0.90-1}{0.25}\right) \\ &= P(Z \le 2.16) - P(Z \le -0.40) \\ &= 0.9846 - 0.3446 \\ &= 0.64. \end{aligned}$$

Example: IQ Scores in a Population

X is the IQ score of a randomly selected individual in a certain population.

$$\{84,\ 127,\ 101,\ 100,\ 131,\ 89,\ 96,\ \ldots,\ 117\}.$$



$$X \sim N(100, 16^2).$$

(a) Find the percentage of the population with an IQ score below 120.

$$\begin{aligned} P(X \le 120) &= P\left(\frac{X-100}{16} \le \frac{120-100}{16}\right) \\ &= P(Z \le 1.25) \\ &= 0.8944. \end{aligned}$$

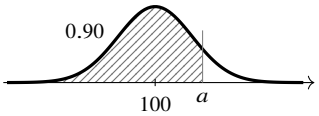
89.44% of the population has an IQ score below 120.

(b) Find the 90th percentile of the IQ scores. (c) Interpret the answer in (b).

$$P(Z \le 1.28) = 0.8997, \qquad P(Z \le 1.29) = 0.9015.$$

Let a denote the 90th percentile, so that

$$P(X \le a) = 0.90,$$
$$P\left(Z \le \frac{a-100}{16}\right) = 0.90.$$



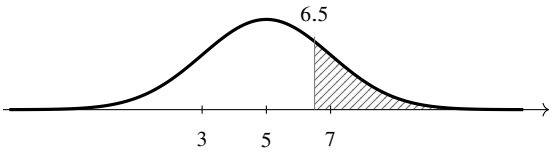
Using 1.28 from Table V,

$$\frac{a-100}{16} \approx 1.28 \quad \Longrightarrow \quad a \approx 100 + 16(1.28) = 120.48.$$

About 90% of the population has an IQ score ≤ 120.48.

Example: Right-Tail Probability

For $X \sim N(5, 4)$, find $P(X > 6.5)$.



$$\begin{aligned} P(X > 6.5) &= 1 - P(X \le 6.5) \qquad \text{(Complement Rule, CR)} \\ &= 1 - P\left(Z \le \frac{6.5 - 5}{2}\right) \\ &= 1 - P(Z \le 0.75) \\ &= 1 - 0.7734 \\ &= 0.2266. \end{aligned}$$

Homework: 40, 42, 44, 46.

Section 4.5: Chebyshev's Rule and the Empirical Rule

Chebyshev's Rule (Any Dataset)

For *any* dataset, at least

$$100\left(1 - \frac{1}{k^2}\right)\%$$

of the values are within k standard deviations of the mean, in the interval

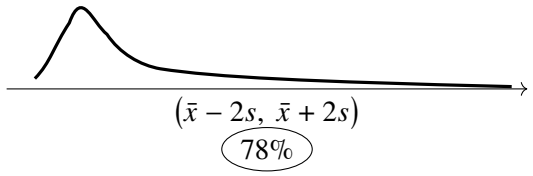
$$\big(\bar{x} - ks,\ \bar{x} + ks\big), \qquad \bar{x} = \text{sample mean}, \quad s = \text{sample standard deviation}.$$

In particular,

$$k = 2: \qquad 100\left(1 - \frac{1}{4}\right)\% = 75\% \text{ (at least)} \qquad (\bar{x} - 2s,\ \bar{x} + 2s),$$

$$k = 3: \qquad 100\left(1 - \frac{1}{9}\right)\% = 88.\overline{8}\% \approx 89\% \qquad (\bar{x} - 3s,\ \bar{x} + 3s).$$

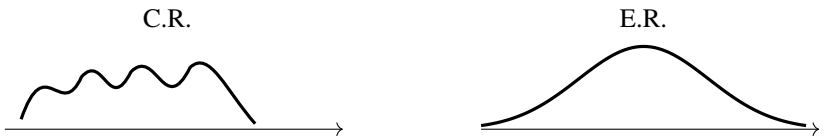
The rule is conservative: the actual percentage can be higher than the lower bound.



The Empirical Rule (Normal Datasets)

For a normal dataset, approximately

- 68% is within 1 standard deviation of the mean;
- 95% is within 2 standard deviations of the mean;
- 99.7% is within 3 standard deviations of the mean.



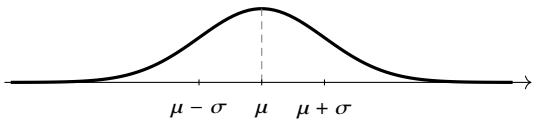
	Within 1 s	Within 2 s	Within 3 s
Chebyshev (any dataset)	—	at least 75%	$\geq 88.\overline{8}\%$ ($\approx 89\%$)
Empirical rule (normal)	$\approx 68\%$	$\approx 95\%$	$\approx 99.7\%$

Chebyshev's rule applies to all datasets and gives lower bounds. The empirical rule gives approximations for normal datasets.

Deriving the Normal-Distribution Percentages

For $X \sim N(\mu, \sigma^2)$, find

$$P(\mu - \sigma < X < \mu + \sigma).$$



$$\begin{aligned} P(\mu - \sigma < X < \mu + \sigma) &= P(X \leq \mu + \sigma) - P(X \leq \mu - \sigma) \\ &= P\left(\frac{X - \mu}{\sigma} \leq \frac{\mu + \sigma - \mu}{\sigma}\right) - P\left(\frac{X - \mu}{\sigma} \leq \frac{\mu - \sigma - \mu}{\sigma}\right) \\ &= P(Z \leq 1) - P(Z \leq -1) \\ &= 0.8413 - 0.1587 \\ &= 0.6826 = 68.26\%. \end{aligned}$$

For two standard deviations,

$$\begin{aligned} P(\mu - 2\sigma < X < \mu + 2\sigma) &= P(Z \leq 2) - P(Z \leq -2) \\ &= 0.9772 - 0.0228 \\ &= 0.9544 = 95.44\%. \end{aligned}$$

For three standard deviations, the entries in Table V give

$$P(-3 < Z < 3) \approx 0.9987 - 0.0013 = 0.9974 \approx 99.7\%.$$

IQ-Score Example: One Standard Deviation

For the IQ-score model $X \sim N(100, 16^2)$,

$$\mu - \sigma = 100 - 16 = 84, \qquad \mu + \sigma = 100 + 16 = 116.$$

Therefore, approximately 68% (more precisely 68.26%) of the population has an IQ score between

84 and 116.

Homework: 48–50; 82, 83, 84, 85, 87(a–d), 88, 91, and 92.

Problem 92: Gamma Function

(a) Show That $\displaystyle\int_0^\infty xe^{-x}\,dx = 1$

Use the gamma-function definition with $\alpha = 2$:

$$\begin{aligned}\int_0^\infty xe^{-x}\,dx &= \int_0^\infty x^{2-1}e^{-x}\,dx \\ &= \Gamma(2) \\ &= (2-1)! \\ &= 1! = 1.\end{aligned}$$

Using

$$\Gamma(n) = (n-1)!, \qquad n = 1, 2, 3, \ldots$$

(b) Find $E[X]$ **for** $p(x) = xe^{-x}$**,** $x > 0$

The pdf vanishes for $x \le 0$. Apply the expected-value integral:

$$\begin{aligned}E[X] &= \int_{-\infty}^\infty x\,p(x)\,dx \\ &= \int_{-\infty}^0 0\,dx + \int_0^\infty x \cdot xe^{-x}\,dx \\ &= 0 + \int_0^\infty x^2e^{-x}\,dx \\ &= \Gamma(3) \\ &= (3-1)! = 2! = 2.\end{aligned}$$